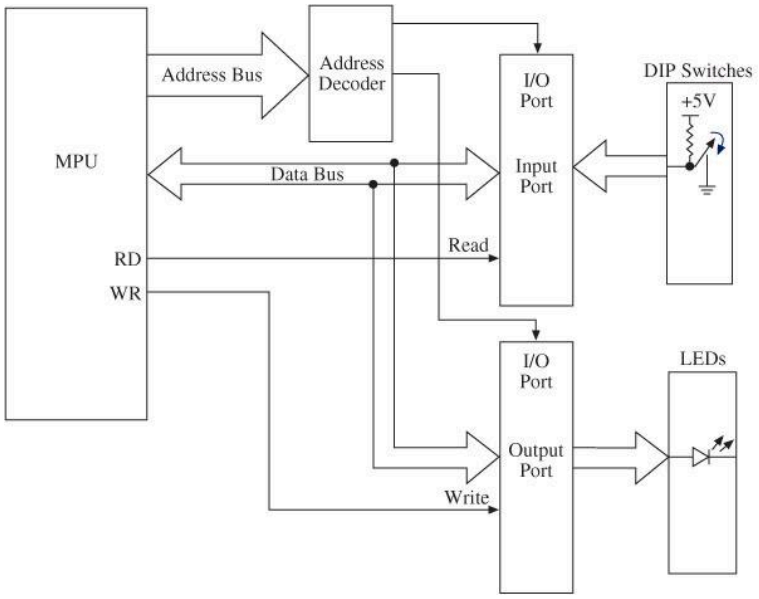
# GPIO

**Input Device:**

1. Dip Switch
2. Push-Button Switch
3. Matrix/Hex Keyboard
4. Analog Input (Sensors)

**Output devices:**

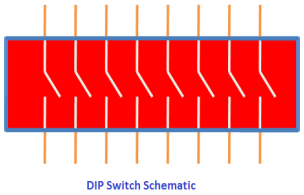
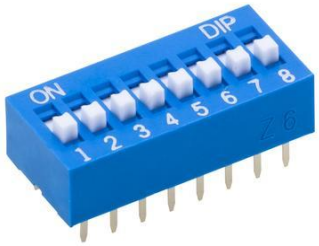
1. 7 Segment display
2. LED Display
3. LCD Display

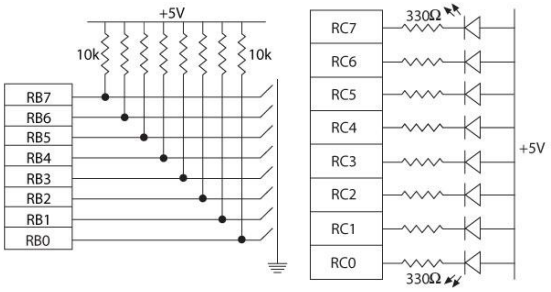



- To read (receive) binary data from an input peripheral
  - MPU places the address of an input port on the address bus, enables the input port by asserting the RD signal, and reads data using the data bus.
- To write (send) binary data to an output peripheral
  - MPU places the address of an output port on the address bus, places data on data bus, and asserts the WR signal to enable the output port.

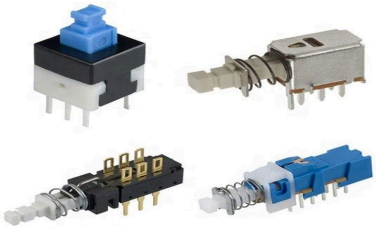
# Input Devices

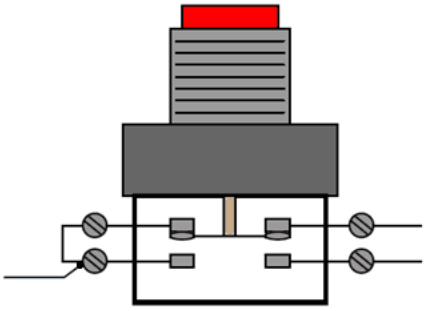
## 1. Dip switch:

| | | |
|---|---|---|
| One side of the switch is tied high (to a power supply through a resistor called a pull-up resistor), and the other side is grounded. The logic level changes when the position is switched. |  DIP Switch Schematic |   |

| | |
|---|---|
|   | Interfacing Dip Switches and Interfacing LEDs follow the same rules. |

## 2. Push-Button switch

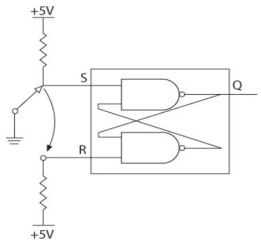
| | |
|---|---|
| The connection is the same as in the DIP switch except that contact is momentary.<br>When a key is pressed (or released), mechanical metal contact bounces momentarily and can be read. |  |



**Note:** Push-Button switches has a problem called Key-debouncing problem

**Key debouncing problem:** .

**Solution of key debouncing problem:**

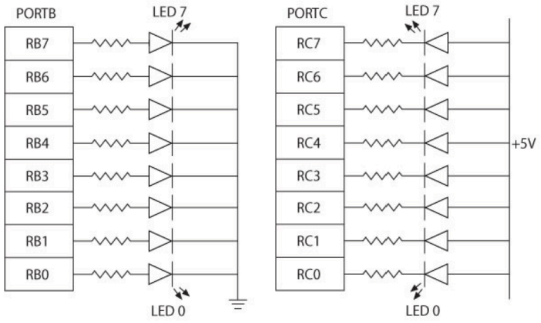
| **Software key debouncing solution:** | **Hardware key debouncing solution:** |
|---|---|
| ● Wait for 20 ms.<br>● Read the port again. | ● Using a S-R latch along with the switch<br>  |

# Output Devices

1. **7 Segment display:**
   - First need to know LED interfacing

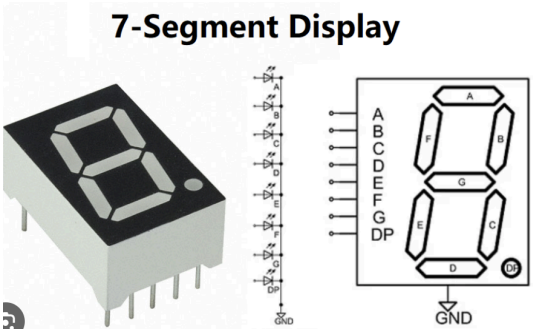
Two ways of connecting LEDs to I/O ports:

- Common Cathode - The current is supplied by the I/O port called current sourcing.
- Common Anode - The current is received by the chip called current sinking.

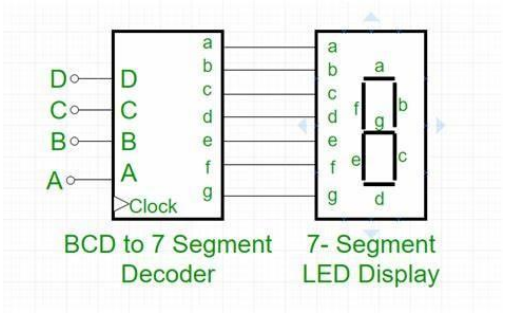



Common Cathode Common Anode

   - 7 segment display:

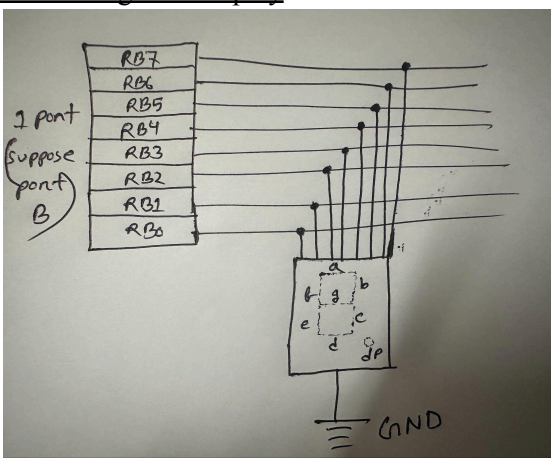
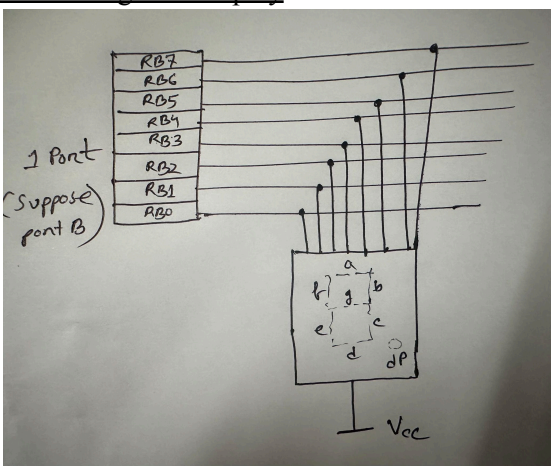


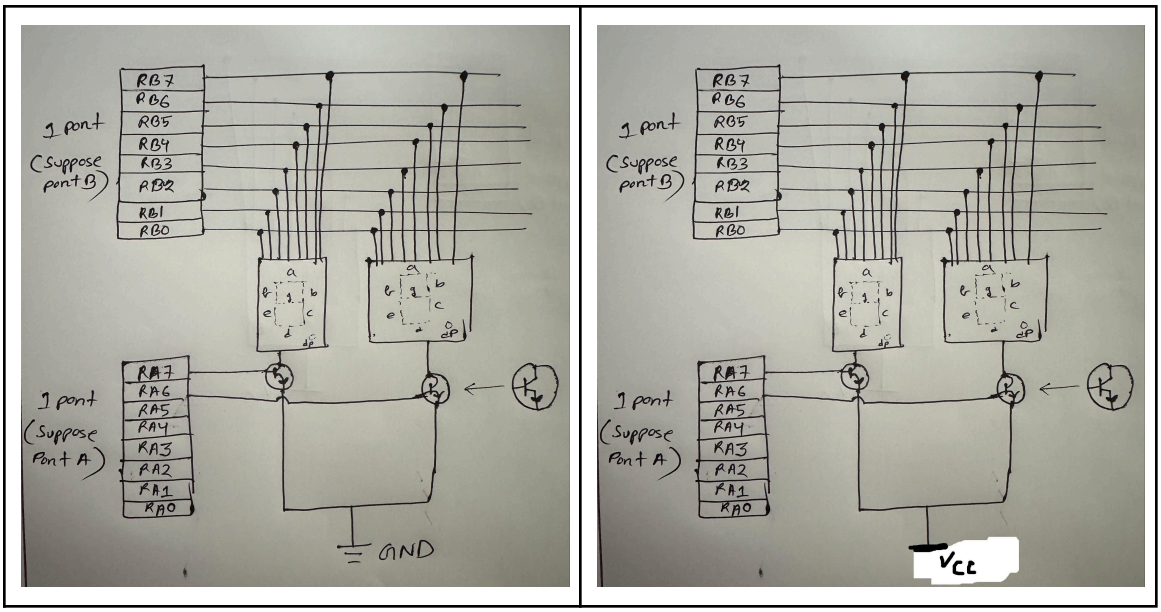
- BCD to 7-Segment Display table: [Common Cathode]



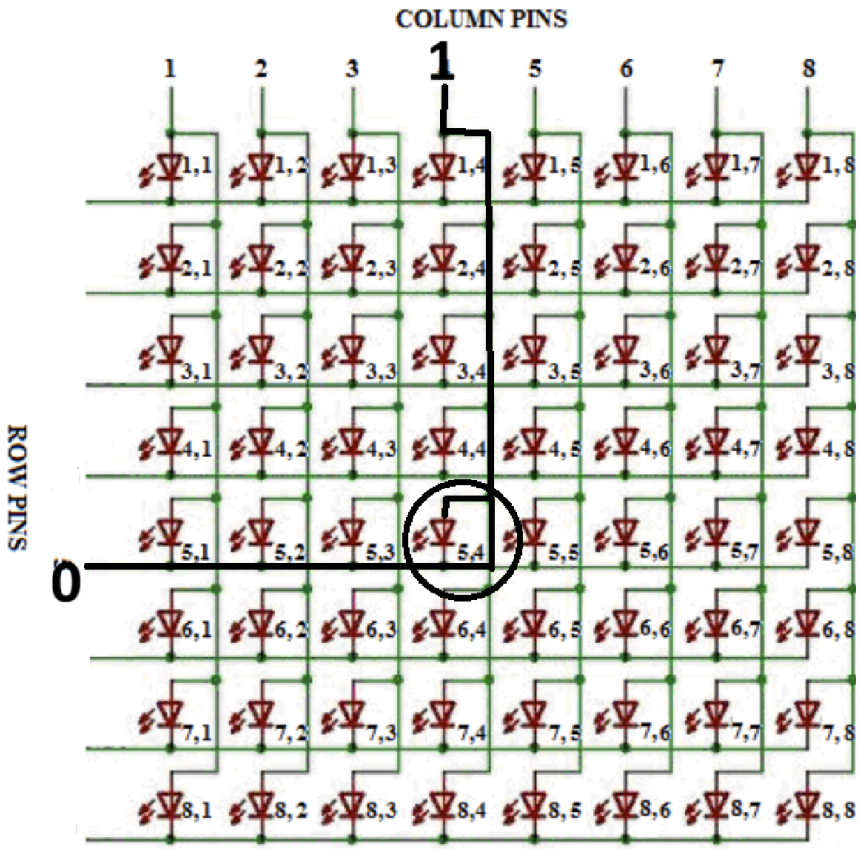

| Decimal Digit | Input lines | | | | Output lines | | | | | | | Display pattern |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | A | B | C | D | a | b | c | d | e | f | g | |
| 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 2 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 2 |
| 3 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 3 |
| 4 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 4 |
| 5 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 5 |
| 6 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 6 |
| 7 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 7 |
| 8 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 8 |
| 9 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 9 |

- 2 ways to interface a 7-segment display: Common Anode & Common cathode

| Common cathode mode | Common Anode mode |
|---|---|
| For 1 7-segment display<br>  | For 1 7-segment display<br>  |
| For multiple 7-segment display (Use transistor for controlling displays) | For multiple 7-segment display (Use transistor for controlling displays): |




2. **<u>LED Matrix Display:</u>**
    - Has 16 pins [Row 8 & column 8]
    - Row pins serves as a sourcing/sinking point and actual data is sent through the column pins
    - If a row pin is grounded [0] and from a column data [1] is sent only that specific LED will lit up. - [common cathode]
    - If a row pin is powered [1] and from a column data [0] is sent only that specific LED will lit up. - [common anode]

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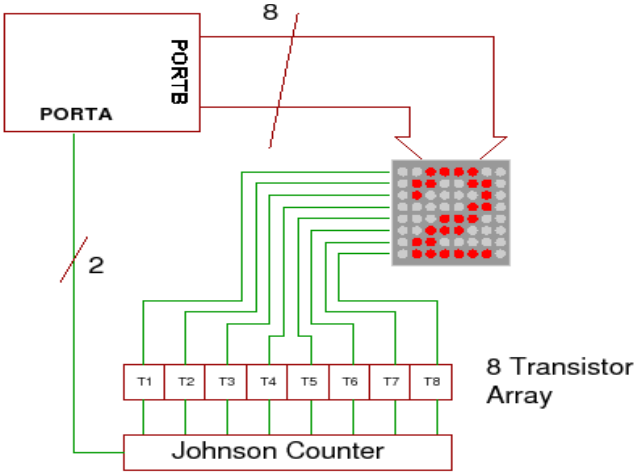
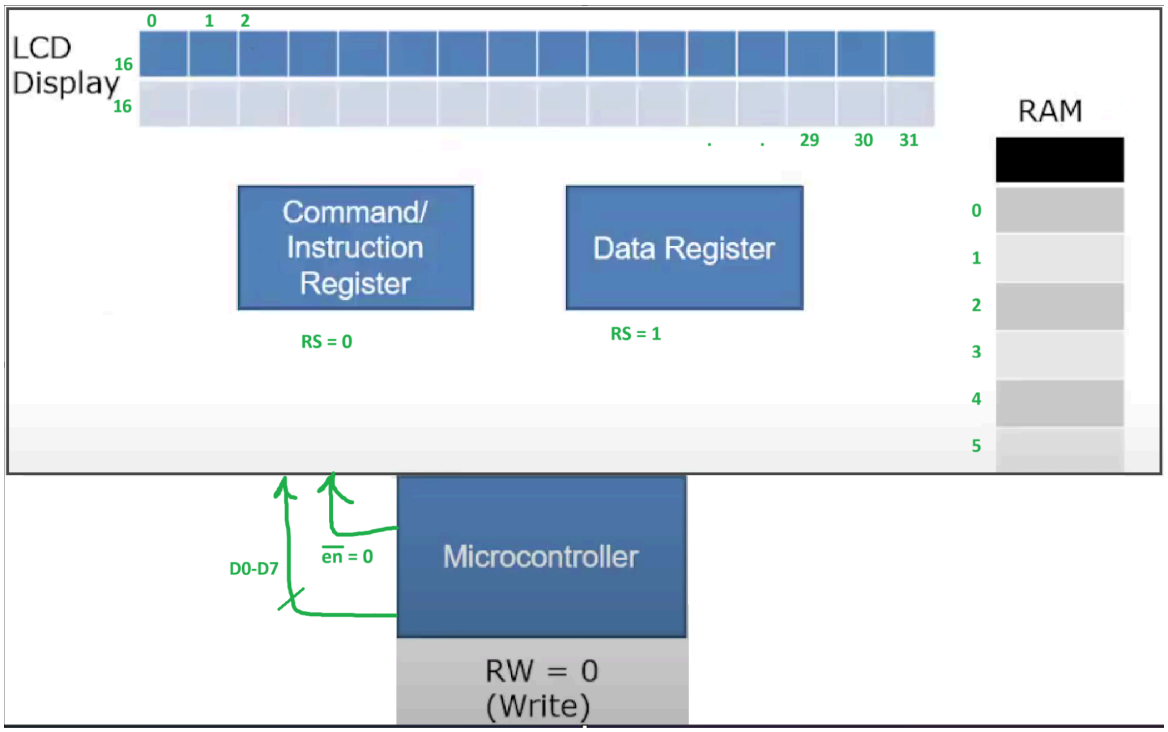
**Note:** 16 pins are too computationally heavy therefore we can use Johnson counters to reduce the pins needed.

- With the Johnson counter only 10 pins are needed.
- Columns will send data as usually [Through 8 pins]
- In the rows we will connect a Johnson counter [Which is basically an array-like structure of 8 transistors]. JC has 2 pins: 1 enable pin and 1 data pin to change clock pulse
- With each clock pulse the data sent [1/0] from the data pin will shift 1 bit and light up the LEDs according to the data sent through the columns.

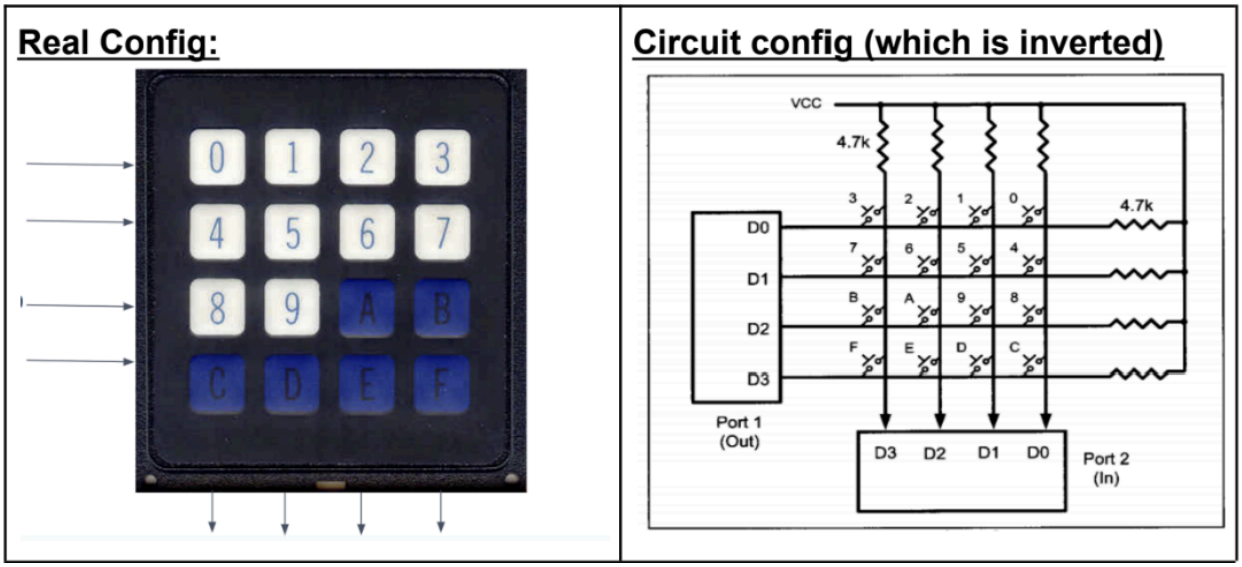


3. **LCD display:**

1. Microcontroller will activate the LCD first [en = 0]
2. Microcontroller will send 0 through D0-D7 which will work to activate the register select pin [RS = 0] to configure the LCD
3. Microcontroller will send 1 through D0-D7 which will work to activate the register select pin [RS = 1] to activate data register
4. Now microcontroller will send the data through D0-D7
5. The data will go to the data RAM through data register
6. Finally the data will be displayed on the LCD from data RAM




**Matrix/ HEX Keyboard**




**How does it work?**

First need to set up the IC

**MOV AL, 82H**
**OUT CWR, AL**

1. **Column identification:**

a. Initially the column values remain 1 since connected to VCC. Meaning, D3-D2-D1-D0 = 1-1-1-1
b. Once any key is pressed the column value is changed to 0. E.g. for pressing "9" the value of D3-D2-D1-D0 = 1-1-0-1
c. The column is identified since the value has changed and it's not 1111 anymore
d. Now this value 1101 (D3-D2-D1-D0 = 1-1-0-1) gets saved in a register for later use [Keypress identification]

```
Row:        MOV AL, 0
            OUT PA, AL

KEYPRESS: IN AL, PB
          CMP AL, 0FH
          JZ KEYPRESS
          CALL DELAY
```

2. **Row identification:**

a. Initially the row values remain 0 from the end of the microprocessor
b. With each pulse microprocessor sends 0 from one pin (at first D0 = 0) and 1 from others (D3-D3-D1 = 1-1-1)

c. With each clock pulse and each change the system searches for the intersecting point which completes the circuit. Once the intersecting point is found the row value has also been identified. For example D2 in this case.

**R0:**

```
MOV AL, 1110
OUT PA, AL
IN AL, PB
CMP AL, 0FH
JZ R1
LEA SI, R0
JMP COL
```

3. **<u>Key press identification:</u>**

a. We got **the value of column 1101** and row ( in this case Row 2 (R2))

b. Now we have to find the 0 by shift right (SHR) operation from the mapping to find out the corresponding value and identify the key finally.

**COL:**

```
MOV AL, [SI]
```